Amanpreet Singh

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Summary

ML focused Software Engineer with 1.5+ years of industry experience at Shell and Vakrangee, and funded graduate researcher in spatial clustering using Transformers at McMaster University. Built production grade data pipelines, RAG systems, and ServiceNow automations that cut operational overhead by 20%+. Looking to bring that end-to-end ML experience to an ML Engineer or Data Scientist role where models solve real problems and impact is measured.

Experience

Data Science Teaching Assistant | McMaster University | Hamilton, Canada — **September 2025 – Present**

- Instructed 60+ graduate students in statistical modeling, machine learning, and data science through weekly labs and structured office hours, contributing to improved course completion rates.
- Graded assignments and provided detailed feedback, reinforcing applied understanding of core data science concepts.

Systems Engineer | Shell India Markets Pvt. Ltd. | Bengaluru, India — **August 2023 – December 2024**

- Responsible for end-to-end automation for 3 critical microservice modules (joiner/mover/leaver), designing REST API integrations and server-side JavaScript workflows that eliminated manual steps for 500+ employee transitions annually.
- Reduced developer rework by 2 weeks per release cycle by leading impact analysis on automated flows, records, and design infrastructure for a ServiceNow platform serving 10,000+ users.
- Drove a 20% reduction in incident resolution effort by building trend analysis dashboards on ServiceNow Incident Management data, identifying recurring root causes and enabling proactive SLA compliance.

Data Analyst | Vakrangee Ltd. | Internship | Mumbai, India — **May 2022 – June 2022**

- Developed views, stored procedures, and implemented CRUD operations on SQL Server Management Studio (SSMS) to optimize database performance and streamline data retrieval, ensuring high availability and reliability of data resources.
- Managed and developed data warehousing solutions by applying ETL techniques to extract, transform, and load data from multiple sources ensuring data consistency and accuracy to support analytics and business reporting on company's website.

Skills

Languages & Programming | Python, SQL, JavaScript, R, C, HTML/CSS, Git, Bash/Shell, NumPy, Pandas, Scikit-learn

AI/ML & Data | PyTorch, TensorFlow/Keras, HuggingFace Transformers, LangChain, LangGraph, MLflow, OpenAI APIs, RAG, Vector Databases, Feature Engineering, ETL Pipelines, Data Warehousing, REST APIs, SSMS, Tableau/Power BI

Tools & Platforms | Azure (AZ-900 Certified), AWS/GCP, Docker, Jupyter, ServiceNow, GitHub, Jira, RStudio

Project Work

AgenticAI | GitHub — **Jan 2026**

- Built an end-to-end business intelligence application using Python, Streamlit, and DeepSeek LLM that automates competitive analysis by orchestrating web research, NLP pipelines, and SWOT synthesis into downloadable professional reports.
- Designed a multi-stage agentic AI pipeline integrating HuggingFace Inference API, DistilBERT, MiniLM embeddings, and KMeans clustering to transform unstructured web data into structured strategic insights with real-time streamed output.

Cost Optimized Credit Risk Modeling | GitHub — **Nov 2025**

- Built a credit risk prediction model achieving strong AUC on loan default classification, incorporating asymmetric loss functions to reduce false-negative approval costs in a cost-sensitive financial framework.
- Tracked experiments and hyperparameter tuning using MLflow, enabling reproducible model evaluation across multiple classifiers.

House Price Prediction | Pipeline | GitHub — **June 2025**

- Deployed an end-to-end ML pipeline (data ingestion → preprocessing → training → API endpoint) for real estate price prediction, with automated retraining via CI/CD on GitHub Actions.

Education

Master's in Computational Science and Engineering | McMaster University | Hamilton, Canada — **December 2026**

Funded research student, developing a nonparametric framework for repeated spatial clustering using Transformer architectures.

Bachelor of Computer Science & Business Systems | NMIMS University | Mumbai, India — **May 2023**

CGPA: 3.70/4

Publications

Audio Classification of Emergency Vehicle Sirens Using Recurrent Neural Network Architectures | Springer Book Series — **Oct 2023**

Review Of Approaches Towards Building AI Based Career Recommender & Guidance Systems | Scienceopen — **August 2024**